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Integrated Neuromorphic Photonic Computing for AI Acceleration: Emerging Devices, Network Architectures, and Future Paradigms

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Deep learning stands as a cornerstone of modern artificial intelligence (AI), revolutionizing fields from computer vision to large language models (LLMs). However, as electronic hardware approaches fundamental physical limits—constrained by transistor scaling challenges, von Neuman architecture, and thermal dissipation—critical bottlenecks emerge in computational density and energy efficiency. To bridge the gap between algorithmic ambition and hardware limitations, photonic neuromorphic computing emerges as a transformative candidate, exploiting light's inherent parallelism, sub-nanosecond latency, and near-zero thermal losses to natively execute matrix operations—the computational backbone of neural networks. Photonic neural networks (PNNs) have achieved influential milestones in AI acceleration, demonstrating single-chip integration of both inference and in situ training—a leap forward with profound implications for next-generation computing. This review synthesizes a decade of progress in PNNs core components, critically analyzing advances in linear synaptic devices, nonlinear neuron devices, and network architectures, summarizing their respective strengths and persistent challenges. Furthermore, application-specific requirements are systematically analyzed for PNN deployment across computational regimes: cloud-scale and edge/client-side AIs. Finally, actionable pathways are outlined for overcoming material- and system-level barriers, emphasizing topology-optimized active/passive devices and advanced packaging strategies. These multidisciplinary advances position PNNs as a paradigm-shifting platform for post-Moore AI hardware.

1. Introduction

In the era of big data and unprecedented computational advances, deep learning[1,2] has emerged as the cornerstone of modern artificial intelligence (AI). By leveraging hierarchical architectures to extract high-level features from raw data, deep learning has revolutionized the way machines learn and perform complex tasks with human-like proficiency. Its distributed structure, enabled by parallelized parameter optimization, confers exceptional fault tolerance and data-driven adaptability, driving breakthroughs across domains such as computer vision,[3,4] natural language processing,[5–7] and probabilistic forecasting.[8] Landmark achievements like AlphaGo's strategic decision-making,[9,10] ChatGPT's generative dialogue,[11,12] Sora's video synthesis,[13,14] and Waymo's autonomous navigation[15] underscores deep learning's transformative role in rendering human-machine collaboration, solidifying its position as the engine of the ongoing AI revolution. Yet, the escalating complexity and scale of deep learning models—from billion-parameter to multimodal foundation architectures—expose fundamental limitations in conventional van Neumann computing paradigms. Modern deep learning architectures are structurally defined by cascaded linear-nonlinear transformations: linear weight matrices (e.g., convolution, fully connected layers, attention mechanisms) perform high-dimensional tensor operations, while nonlinear activation modules (e.g. ReLU, GELU, or stochastic regularization layers) introduce critical decision boundaries. These operations demand extensive computational intensity. The significant computational intensity of these operations exposes the limitations of conventional von Neumann computing, which is constrained by the memory wall (due to physical separation of processing and memory units) and limited parallelism.[16] To address these challenges, industry leaders have developed specialized hardware accelerators tailored to the computational patterns of deep learning, including Graphics Processing Units (GPUs),[17] Tensor Processing Units (TPUs)[19,20] and Field Programmable Gate Arrays (FPGAs).[21,22] While mitigating the von Neumann bottleneck through task-specific parallelism and

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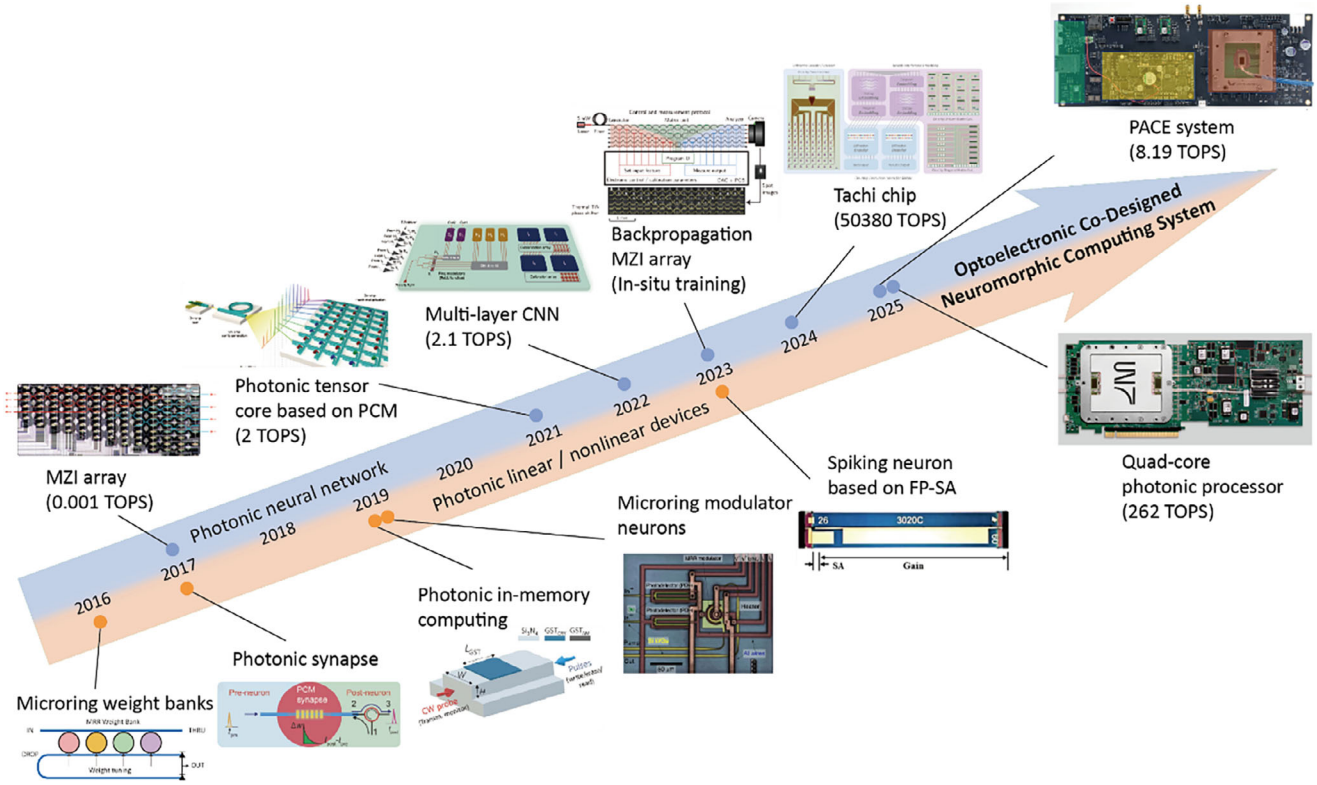


Figure 1. Evolution of integrated photonic neuromorphic computing. Major milestones in the development of photonic neuromorphic devices and neural networks are illustrated from 2016 to 2025. In the domain of photonic linear/nonlinear devices, the 2016 demonstration of microring weight banks laid the groundwork for subsequent progress. This foundation led to the emergence of various photonic synapse and neuron devices, including photonic synapses based on PCMs and photonic neuron implementations using microring modulators and FP-SAs. For photonic neural networks, the 2017 publication of MZI arrays marked a pivotal milestone, sparking continuous progress in both network complexity and computational throughput. Since then, photonic chips have evolved from single-layer to multi-layer neural networks, while their compute performance has steadily increased. This trajectory culminated in the 2024 Tachi chip, which achieved a record-breaking performance exceeding 50,000 TOPS—demonstrating remarkable advancements in integration scale and parallel processing capability. In addition, recent advances (e.g., in 2023) have enabled on-chip training through in situ backpropagation, suggesting that PNNs are becoming increasingly autonomous and capable of real-time learning and adaptation in dynamic environments. Notably, emerging systems like the PACE (photonic arithmetic computing engine) system and the quad-core photonic processor signal a paradigm shift toward optoelectronic co-design for next-generation neuromorphic computing, merging photonic speed with electronic programmability. 'Photonic synapse': Reproduced with permission.[57] Copyright 2017, AAAS. 'Photonic in-memory computing': Reproduced with permission.[58] Copyright 2019, AAAS. 'Microring modulator neurons': Reproduced with permission.[39] Copyright 2019, American Physical Society. 'Spiking neuron based on FP-SA': Reproduced with permission.[52] Copyright 2023, Optica Publishing Group. 'MZI array': Reproduced with permission.[38] Copyright 2017, Springer Nature. 'Photonic tensor core based on PCM': Reproduced with permission.[49] Copyright 2021, Springer Nature. 'Multi-layer CNN': Reproduced with permission.[41] Copyright 2022, Springer Nature. 'Backpropagation MZI array (In-suit training)': Reproduced with permission.[43] Copyright 2023, AAAS. 'Taichi chip': Reproduced with permission.[45] Copyright 2024, AAAS. 'PACE system': Reproduced with permission.[59] Copyright 2025, Springer Nature. 'Quad-core photonic processor': Reproduced with permission.[60] Copyright 2025, Springer Nature.

near-memory compute, these electronic solutions face inherent trade-offs. For instance, resistance-capacitance time delays[23] impose a hard ceiling on signal processing speeds, restricting clock frequency scaling even at advanced nodes (e.g. ≈5 GHz maximum in 5 nm CMOS[24]). Concurrently, electromagnetic crosstalk[25] between densely packed interconnects degrades signal integrity, exacerbating timing errors and limiting computational precision, particularly in analog neuromorphic architectures. These challenges underscore the need for innovative approaches to hardware architecture for deep learning applications.

Harnessing the broadband spectrum utilization and wavelength-division multiplexing (WDM) capabilities, photonic computing[26] offers revolutionary advantages for AI acceleration: terahertz-bandwidth parallelism,[27] ultralow-latency signal propagation, and inherent immunity to electromagnetic crosstalk. These properties enable photonic neural networks (PNNs) to natively execute core AI operations—such as matrix multiplications and convolutions—with pJ/operation energy efficiencies, bypassing the resistance-capacitance delay bottlenecks and thermal noise constraints of electronic counterparts. This positions photonic computing as a promising solution for post-Moore AI hardware, capable of scaling to zettascale computational demands while sustaining energy-efficient, real-time inference in applications ranging from hyperscale data centers to autonomous edge systems. In the past decades, foundational advancements in integrated photonics have propelled optical components toward high-speed operation, ultralow power consumption, and sub-wavelength miniaturization. Breakthroughs in key building

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blocks—such as heterogeneously integrated III-V/Silicon light sources,[28] ultracompact electro-optic modulators,[29,30] dynamically tunable optical attenuators,[31] low-noise semiconductor optical amplifiers,[32] high-fidelity wavelength-division (de)multiplexers,[33] and high-speed photodetectors[34,35]—have matured the toolkit for photonic integrated circuits (PICs). Coupled with advances in nonvolatile photonic memories,[36] these innovations now enable the co-design of monolithic PNN chips capable of CMOS-compatible foundry production. While full-stack optical implementation of deep learning workflows remains an open challenge, state-of-the-art PNNs already integrate linear synaptic weight arrays and nonlinear neuron modules on a single chip. These architectures achieve groundbreaking computational densities (>10 trillion of operations per second per square millimeter, TOPS/mm^2) and energy efficiencies (<100 fJ/operation)—outperforming electronic counterparts by 1–3 orders of magnitude in matrix-intensive tasks.

The development of on-chip PNNs traces its roots to advancements in silicon photonics, a platform that integrates optical components with CMOS-compatible fabrication processes. Over the past decade, this synergy has driven the rapid evolution of neuromorphic photonic devices and networks, as summarized in **Figure 1**. Early milestones include the 2016 demonstration of micro-ring resonator (MRR) weight banks,[37] which exploited wavelength-selective resonance to emulate tunable synaptic weights—a foundational step toward optical matrix operations. A transformative leap occurred with the pioneering work of Shen et al., who introduced a programmable nanophotonic processor using cascaded Mach–Zehnder interferometers (MZIs).[38] By leveraging the phase-dependent interference of light in MZI meshes, this architecture enabled reconfigurable linear transformations, laying the groundwork for photonic acceleration of deep learning workloads. Subsequent innovations addressed the critical need for nonlinear activation functions, a cornerstone of neural network learning. Breakthroughs such as microring modulator neurons[39] (utilizing resonance shifts for thresholding) and photocurrent-modulated MZI neurons[40] (combining optoelectronic feedback with phase tuning), bridged the gap between linear optics and nonlinear computation. These advances unlocked the implementation of large-scale PNNs. The field reached a watershed moment with the monolithic integration of multi-layer deep PNNs.[41,42] These architectures demonstrated end-to-end inference for tasks like image classification, executing photonic convolution and nonlinear activations entirely on-chip. Critical advances in photonic training were realized through in situ learning frameworks for large-scale PNNs, enabled by optical-power-monitored MZI arrays[43] and tunable diffraction units.[44] Very recently, a landmark achievement was demonstrated by Tsinghua University's Taichi chips,[45] which achieved an unprecedented energy efficiency to 160 TOPS/W. This represents a three-order-of-magnitude improvement (10^3 ×) over NVIDIA's H100 GPU in energy efficiency. Meanwhile, the exploration of advanced materials such as phase-change materials (PCMs) and heterogeneously integrated devices has catalyzed a paradigm shift in PNN architecture.[46,47] The first all-optical spiking neuron network (SNN)[48] was established, combining PCM-based synapses with wavelength-division multiplexed MRRs, which has showcased strong scalability and self-learning ability in both supervised and unsupervised learning. Additionally, the photonic tensor core,[49] integrating PCMs with a crossbar array architecture, demonstrates enhanced computational density and energy efficiency for 5-bit resolution weight operations in experimental implementations. The successful monolithic integration of PCMs with silicon photonics within the back-end-of-line (BEOL) process underscores the transformative potential of heterogeneously integrated architectures for high-performance PNN chips.[50,51] Besides, III–V lasers[52,53] have emerged as a promising platform for photonic SNNs, leveraging their inherent ultrafast gain dynamics to replicate key neurobiological behaviors. These devices exhibit sub-picosecond response times and sub-femtojoule spike energy thresholds under optical injection locking-metrics that directly emulate biological neurons' dynamics. Recent studies have also explored photonic implementations of reservoir computing (RC)[54,55] and Ising machines,[56] which harness the rich spatiotemporal dynamics of optical systems for efficient temporal processing and combinatorial optimization, respectively. These paradigms complement neuromorphic computing strategies, broadening the design landscape for photonic intelligence. Furthermore, by combining PNN chips with optical connections and communication techniques, these systems could allow instant, seamless interaction between cloud services and edge devices—entirely using light. This eliminates the need to convert signals between optical and electrical domains, slashing delays and energy waste from those conversions.

In this review, we systematically examine the basic structures, operational mechanisms, and performance benchmarks of emerging photonic neuromorphic devices. These include linear weight matrix units and nonlinear activation units for artificial neural networks (ANNs), as well as biomimetic synapses and neurons for biologically inspired neural networks (e.g., SNNs). We critically compare the performance advantages and practical limitations of major photonic neural network paradigms, along with advancements in photonic in situ training implementations. Furthermore, we analyze application-specific requirements for photonic neural chips in both cloud computing and edge/client environment, addressing their distinct computational and training demands. Building on this foundation, we explore the role of active/passive device optimization and optoelectronic co-design strategies in unlocking the transformative potential of neuromorphic photonics for high-speed, energy-efficient intelligent computing.

2. Integrated Neuromorphic Photonic Linear Devices (Synapses)

Linear operations constitute the computational backbone of PNNs, with distinct manifestations across network architectures. In multilayer perceptrons (MLPs), linear operations typically refer to matrix-vector multiplications, equivalent to cascaded multiply-accumulate (MAC) operations. Convolutional neural networks (CNNs) exhibit dual linear processing regimes: 1) Fully connected layers that share architectural similarities with MLP implementations, and 2) Convolutional layers requiring high-dimensional parameter parallelism. Although individual convolutional kernels exhibit compact spatial dimensions (e.g., 3 × 3 or 5 × 5 matrices), the requirement for parallel deployment of hundreds of such kernels across input channels im-

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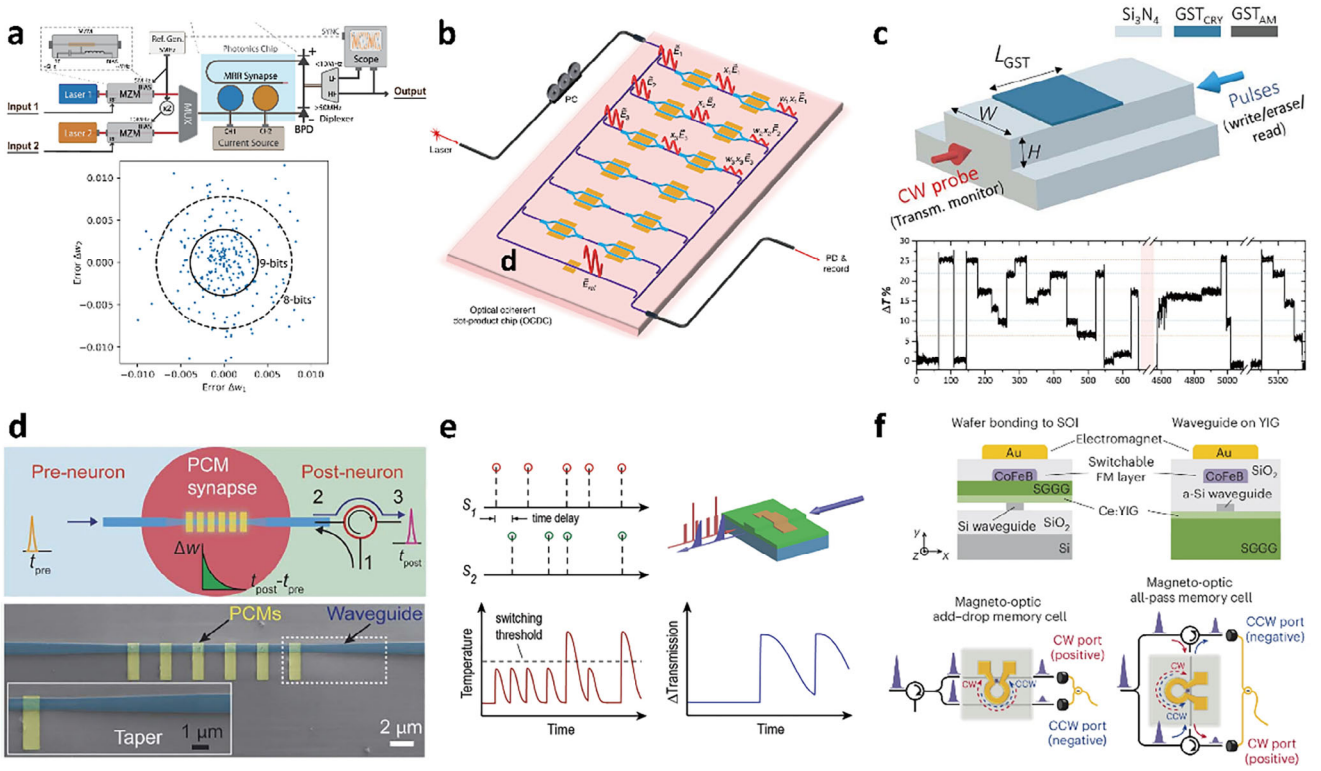


Figure 2. Integrated photonic weight and synapse. a) 9-bit resolution weighted addition using MRRs. The diagram (top) shows the control system utilizing reference signal and feedback.[62] b) Optical coherent dot-product chip that uses push-pull configured MZIs to load weight and input.[64] c) GST-based photonic in-memory computing cell. The diagram (top) shows the scalar multiplication of inputs (light amplitude) and weights (GST state).[58] d) GST-based on-chip photonic synapse allows one to set any weight randomly by varying the optical pulse number. The diagram (top) shows PCM-synapse emulating the STDP dynamic.[57] e) Tunable volatility of GST cell to achieve STP and LTP. The volatility of GST can be regulated by changing the optical pulse interval.[68] f) Non-reciprocal magneto-optical memory device.[69] Schematic cross-sectional diagram (upper section). Utilizing MRR-based non-reciprocal magneto-optical devices, clockwise (CW) and counterclockwise (CCW) optical transmission directions to achieve positive and negative weights (lower section). (a) Reproduced with permission.[62] Copyright 2022, Optica Publishing. (b) Reproduced under the terms of the CC-BY license.[64] Copyright 2021,Springer Nature. (c) Reproduced with permission.[58] Copyright 2019, AAAS. (d) Reproducedwith permission.[57] Copyright 2017, AAAS. (e) Reproduced with permission.[68] Copyright 2019, Wiley. (f) Reproduced with permission.[69] Copyright 2024, Springer Nature.

poses stringent demands on the high-throughput parallel processing architectures of synaptic weight modules in neuromorphic systems. In bio-inspired neural networks, such as SNNs, synaptic modules must not only execute MAC operations but also maintain biological fidelity through dynamic weight adaptation mechanisms. These include spike-timing-dependent plasticity (STDP), short-/long-term potentiation (STP/LTP), and short-/long-term depression (STD/LTD)—critical for temporal coding and neuromorphic learning. In this section, we examine photonic weight implementations—including tunable MRRs, MZIs, semiconductor optical amplifiers (SOAs), and PCMs—alongside synaptic architectures incorporating STDP, STP, and LTP functionalities.

MRRs operate through wavelength-selective resonance in a closed-loop waveguide coupled to a bus waveguide. They feature a through-port (transmits non-resonant wavelengths) and a drop-port (extracts resonant wavelengths), with their spectral response governed by the critical coupling condition. The Q-factor (quality factor) and free spectral range (FSR) of MRRs determine their efficacy in PNNs for weighting and routing operations, with dynamic tuning achieved via thermo-optic (TO) or electro-optic modulation. A balanced photodetector interfaced with these ports enables signed weight representation through differential photocurrent measurement, where the relative intensity between through-port and drop-port signals encodes weight polarity (+/-) in PNNs.[61] MRRs have the advantages of a compact footprint and low power consumption. However, their acute sensitivity to thermal drift and fabrication variations induces phase errors that constrain weight precision. As shown in **Figure 2a**, Zhang et al. demonstrated a closed-loop calibration architecture leveraging MHz dithering signals, achieving 9-bit weight precision.[62] This methodology further enables full-system error calibration. Tait et al. developed a feedback weight control approach based on the circulating power sensing using N-doped heaters,[63] where each MRR can be independently calibrated by monitoring the in-ring resistance, eliminating auxiliary photonic/electronic circuitry. This method achieved a 5-bit weight precision and is suitable for large-scale MRR array calibration. The inherent wavelength sensitivity of MRRs necessitates WDM for multi-input linear operations, requiring multiple wavelength channels matched to their FSR. This spectral specificity introduces scalability challenges for large-scale matrix-vector multiplication (MVM, e.g., fully connected layers in MLPs) due to cumulative phase errors and channel crosstalk. Consequently, MRRs are predominantly

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deployed in CNN architectures, where the smaller size of convolution kernels is more compatible with their functionality. On the other hand, MZIs modulate optical intensity via coherent interference between phase-shifted light in dual waveguide arms. Two primary modulation mechanisms are employed: TO tuning and carrier dispersion. TO-based MZIs dominate photonic computing architectures due to their superior low insertion loss, despite slower switching speeds. As shown in Figure 2b, Xu et al. implemented a photonic AUTOMAP (a cutting-edge neural network model for image reconstruction) image reconstruction neural network[64] by multiplexing a coherent MZI network, achieving an error of less than 0.0028, matching the precision of a 32-bit digital computer. MZIs enable high-precision weight modulation via TO phase shifters by refractive index control. While TO modulation exhibits inherent nonlinearity, this can be effectively compensated through linearization techniques. However, TO-based devices typically require relatively large footprints (≈300 µm) in passive components to accommodate thermal diffusion timescales and minimize inter-element crosstalk. Owing to their excellent parallel processing capabilities, MZI networks can be effectively utilized in a wide range of MLP and CNN architectures.

Beyond weight modulation via optical phase control in MRRs and MZIs, alternative methods leveraging optical amplification or absorption mechanisms have also been explored., Shi et al. demonstrated heterogeneously integrated InP semiconductor optical amplifier (SOA) as photonic weights, achieving a dynamic range of 27 dB.[65] They implemented an 8-element InP photonic network to perform multiply-accumulate operations, achieving 85% classification accuracy on the Iris flower dataset. However, they necessitate heterogeneously integrated InP layers, making the fabrication process more complex. The operational principle of electro-absorption modulators (EAMs) is based on modifying the material's band structure through an external electric field to alter its optical absorption. EAMs offer advantages such as a compact footprint and high modulation speed. Liu et al. developed an EAM featuring epitaxial triangular germanium on a silicon waveguide,[66] achieving a 3 dB bandwidth of 36 GHz with a device footprint of merely 40 × 15 µm. However, the operational wavelength range of EAMs remains relatively limited due to the material's intrinsic absorption wavelength constraints.

Most existing weight modulation schemes rely on volatile mechanisms, necessitating continuous power expenditure to sustain the weight state, which is not favorable for energy-efficient neuromorphic applications. Therefore, non-volatile weight modulation mechanisms—leveraging novel materials—have gained prominence. PCMs exhibit significant changes in refractive index and absorption coefficient between their crystalline and amorphous states. Their non-volatile nature makes them suitable for zero-static-power photonic weights. Ríos et al. demonstrated an in-memory optical multiplication (Figure 2c) by leveraging the transmittance of the $Ge_2Sb_2Te_5$ (GST, a typical PCM) as non-volatile weight.[58] By depositing GST cells on both sides of a Y-shaped waveguide and modulating their transmittance independently, the multiply-accumulate operation for two vectors can be achieved. Li et al. optimized the optical pulses for the crystallization and amorphization processes of PCMs, with improved modulation states of 34 levels (>5 bits).[67] PCMs, with the compact device structure and a broad band working wavelength, are well-suited for parallel computation of multi-channel convolution kernels in CNN architectures. Cheng et al. developed a discrete GST structure integrated with a tapered waveguide shown in Figure 2d, and verified that the transmittance change during the crystallization process exhibits an exponential decay relationship with the number of optical pulses.[57] This behavior is akin to the synaptic plasticity of STDP in neurology. Youngblood et al. demonstrated that the amorphous state retention time of GST decreases exponentially with probe light intensity.[68] By tuning the probe intensity, GST switches dynamically between non-volatile and volatile regimes. As shown in Figure 2e, increasing the temporal separation between amorphization pulses induces a non-volatile-to-volatile transition, with relaxation times gradually decreasing. This behavior aligns with the biological characteristics of synapses having both long-term and short-term memory. Photonic synapses exhibiting characteristics such as STDP and both long-term and short-term memory are highly suitable for neural networks like SNNs. Beyond PCMs, magnetic and ferroelectric materials offer alternative pathways to non-volatile photonics modulation. As shown in Figure 2f, Pintus et al.[69] heterogeneously integrated cerium-substituted yttrium iron garnet (Ce:YIG) with silicon MRRs, achieving GHz-speed non-volatile/volatile magnetic state control via current or RF signals, enabling non-reciprocal magneto-optical effects and endurance exceeding 2.4 billion cycles. In parallel, Geler-Kremer et al.[70] demonstrated a $BaTiO_3$-on-silicon ferroelectric phase shifter, programming eight stable non-volatile states with electrical pulses while delivering ultra-low loss (0.07 dB) and energy efficiency (4.6–26.7 pJ/switch).

3. Integrated Neuromorphic Photonic Nonlinear Devices (Neurons)

Nonlinear activation functions (e.g. ReLU, sigmoid) serve as critical differentiators in ANNs, breaking the symmetry of linear transformations to enable universal approximation of high-dimensional non-convex functions (per Cybenko's theorem).[71] Without nonlinearities, a sequence of linear connections is mathematically equivalent to a single linear layer,[72] underscoring their necessity for capturing complex data topologies through stacked linear-nonlinear processing stages. Neuromorphic photonic systems exploit all-optical and optoelectronic nonlinear responses of materials/devices to implement intensity-dependent or phase-modulated transformations, dynamically mapping input optical signals to nonlinear outputs. These mechanisms emulate nonlinear activation functions in photonic neural networks. Besides, photonic neurons must achieve integration of weighted optical signals (fan-in) through coherent interference or photodetection-mediated summation, while maintaining bidirectional cascadability (fan-out)—output signals must preserve phase/amplitude fidelity to drive connected neurons.[39,73] Nonlinear activation in conventional ANNs operates via clock-synchronized continuous nonlinearities, whereas SNNs require event-driven nonlinear dynamics implementing leaky integrate-and-fire and spatiotemporal integration[74–76] as well as stochastic dynamics.[77,78] In many PNN chips, nonlinear neurons are the primary bottleneck of the operating speed, integration density, and energy consumption of the whole system.[41,48]

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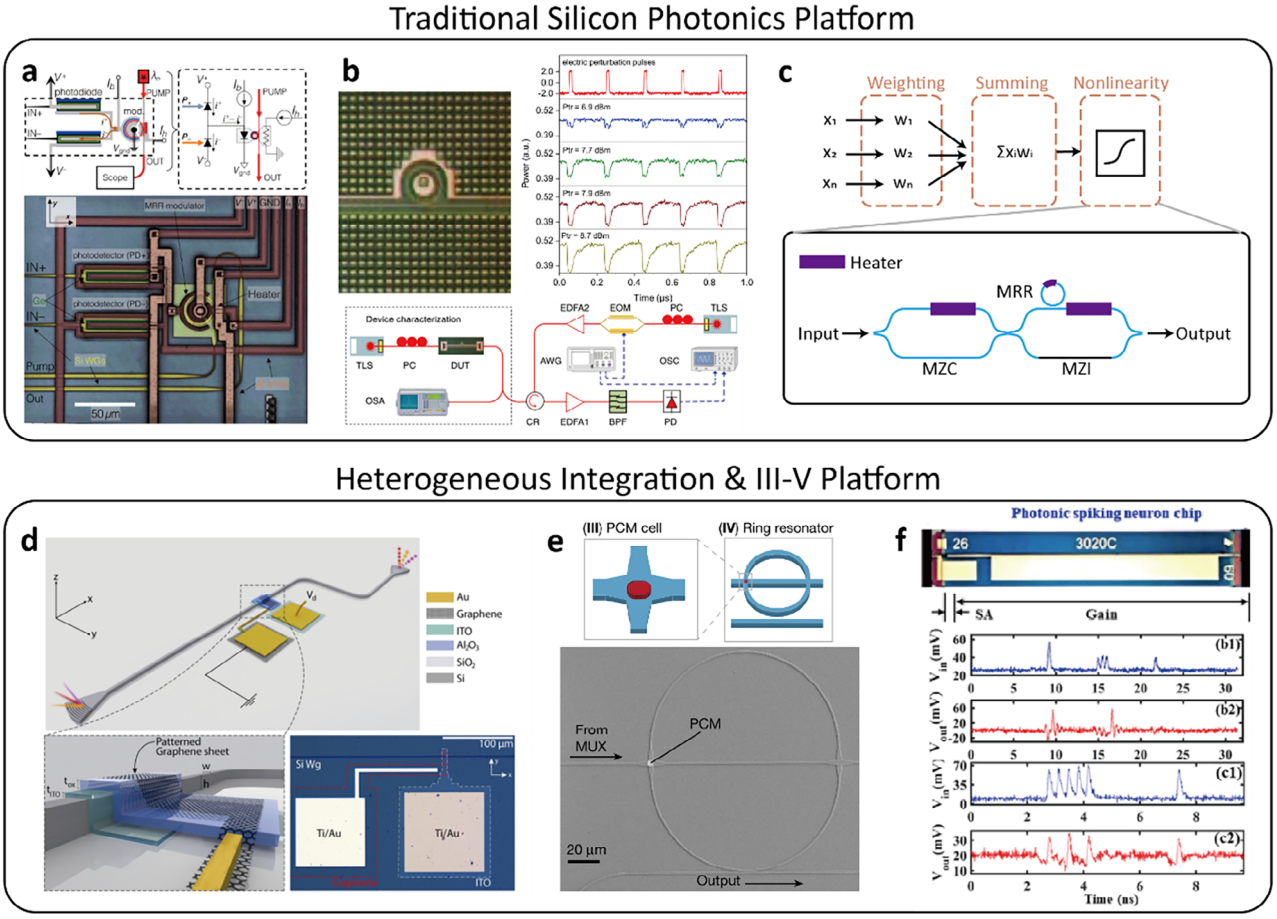


Figure 3. Integrated photonic nonlinear neurons. a), Silicon MRR modulator neuron.[39] The excitatory/inhibitory weighted summation is converted to the photocurrent from the positive/negative photodetector and adds to/shunts away the bias current of the microring modulator, tuning the electro-optic nonlinear transfer function. b), All-optical silicon microring spiking neuron.[81] Subthreshold "negative" spikes emerge under the condition of optical perturbation power exceeding a critical value but remaining below a narrow threshold region. Once perturbation power surpasses this threshold, spike strength saturates, resembling a class II resonate-and-fire neuron. c), Cavity-loaded MZI neuron realizing reconfigurable nonlinear activation functions with tunable thresholds.[79] Thermo-optic heaters on MRR, MZI, and Mach–Zehnder coupler (MZC) tune amplitudes and phase biases to program nonlinear transfer functions. d), Electro-absorption modulator (EAM) based on ITO–graphene heterojunction integrated with silicon photonic waveguides.[83] The diode-like nonlinear response of the device functions as a leaky rectifying linear unit (ReLU). e), All-optical PCM-based neuron.[48] The probe (bus) waveguide produces output spikes when the summed power of input pulses from the multiplexer is high enough to switch the PCM cell on the ring resonator. f), Photonic spiking neuron based on integrated Fabry–Perot laser with a saturable absorber[52] can be triggered by small external input light perturbations to generate output spikes in the condition of the bias current and reverse voltage set below the Q-switching pulse threshold. a) Reproduced with permission.[39] Copyright 2019, American Physical Society. b) Reproduced with permission.[81] Copyright 2022, Chinese Laser Press. c) Reproduced with permission.[79] Copyright 2020, Optica Publishing Group. d) Reproduced with permission.[83] Copyright 2021, AIP Publishing. e) Reproduced with permission.[48] Copyright 2019, Springer Nature. f) Reproduced with permission.[52] Copyright 2023, Optica Publishing Group.

State-of-the-art photonic nonlinear neurons are predominantly developed on Si/Si_3N_4 waveguide platforms, leveraging their mature foundry processes. A widely adopted approach exploits the inherent nonlinear properties of Si/Si_3N_4 photonic devices, such as MRRs[39,41] and MZIs.[40,79] As illustrated in **Figure 3a**, silicon photonic microring modulator neurons—first demonstrated in[43]—leverage the ultrafast electro-optic free carrier dispersion effect to achieve neuron activation with sub-ns speed, constrained primarily by photodetector bandwidth. These devices have been monolithically integrated into large-scale PNN chips,[41] demonstrating their potential for practical applications. While thermo-optic effects exhibit limited response speeds and high power consumption,[80] thermo-optical microring neurons[81] in Figure 3b are capable of biomimetic emulation of spiking dynamics of biological neurons, making them suitable for photonic SNN applications. MZI neurons are implemented via two primary architectures: photocurrent–modulated MZI neurons,[40] where a fraction of the optical input is converted to analog electrical signals to drive phase shifters for nonlinear activation, and all-optical phase-tunable cavity-loaded MZI neurons[79] (Figure 3c), which employ programmable phase shifters to tailor nonlinear transfer functions (e.g. Sigmoid, Softplus, ReLU) with energy costs dominated by static phase maintenance. The primary challenge for Si/Si_3N_4 waveguide photonic neurons lies in footprint and power consumption reduction to satisfy large-scale integration.

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Table 1. Comparison of performance metrics for various integrated neuromorphic photonic nonlinear devices.

Structure	Signal domain	Platform	Size [μm^2]	Operation speed [ps/op]	Consumption [pJ/op]	Monolithic integrability	Mechanism
BPD + MRR[39]	Electro-optic	Si	415.48	500	–	Yes	FCD
BPD + MRR[41]	Electro-optic	Si	201	46	–	Yes	FCD
PN + MDM[92]	Electro-optic / All-optical	Si	314	1000	–	Yes	Free carrier injection
FP-SA[52,89]	Electro-optic	InP	4 × 5 10^5	300	3.248 × 10^{-2}	No	Saturable absorption
Quantum-dot laser[53]	Electro-optic	GaAs	12000	4000	4 × 10^{-2}	No	Quantum dot gain
VCSEL[91]	Electro-optic	GaAs	900	1000	2.5 × 10^{-3}	No	Homodyne interference
ITO/Al_2O_3/Graphene[83]	Electro-optic	Si	5	7.69	2.43	Yes	Absorption modulation
ITO/SiO_2/Au[82]	Electro-optic	Si	4	40	0.06	Yes	Electro-absorption modulation
MZC+MZI+MRR[79]	All-optical	Si	18184	1000	–	Yes	FCD
MRR[81]	All-optical	Si	154	15000	12	Yes	FCD and TO effect
MRR[80]	All-optical	Si	314	1.2 × 10^7	960	Yes	TO effect
Mxene on waveguide[90]	All-optical	Si	600	–	–	No	Absorption modulation
GST on waveguide[85]	All-optical	Si	0.5	100	200	Yes	TO effect of GST
Ge on MRR[86]	All-optical	Si	470.47	10^7	7400	Yes	Absorption modulation
GST on MRR[48]	All-optical	Si_3N_4	11310	2 × 10^5	430	Yes	TO effect of GST

BPD, balanced photodiodes; MRR, microring resonator; PN, PN junction; MDM, microdisk modulator; FP-SA, Fabry–Perot laser with a saturable absorber; VCSEL, vertical-cavity surface-emitting laser; MZC, Mach–Zehnder coupler; MZI, Mach–Zehnder interferometer; Mxene, 2D metal carbides and nitrides; GST, $Ge_2Sb_2Te_5$; FCD, free carrier dispersion; TO, thermo-optical.

To transcend the performance limits of Si/Si_3N_4 neurons, heterogeneously integrated neurons combine PCMs, 2D materials, and III–V semiconductors on Si/Si_3N_4 platforms. This multi-material synergy achieves higher modulation efficiency, faster response speeds, and lower power consumption via monolithic or hybrid integration techniques. Particularly, PCMs and 2D materials integrated on Si/Si_3N_4 platforms achieve ultra-compact neuron designs (<5 μm^2) without added vertical thickness, ensuring compatibility with future 3D-stacked PNN architectures. Single indium tin oxide (ITO) layers[82] and ITO–graphene heterojunctions[83] (Figure 3d) integrated on waveguides function as nonlinear electro-optic absorption modulators. They convert weighted summation inputs into electrical signals to directly modulate optical outputs. GST, a representative PCM, enables programmable nonlinear functions[84] when integrated on MRRs, operating at nJ-level energy costs for PNN implementations[48] (Figure 3e). However, GST-based neurons require post-operation reset to their initial states, fundamentally limiting switching speeds. While GST/Si waveguides demonstrate 25 ns volatile responses,[85] achieving ultra-fast volatile PCMs with low optical switching thresholds remains unresolved. Parallel efforts explore low-threshold thermal-optical materials like germanium (Ge)[86] for programmable nonlinear activations with reduced reset overhead.

III-V laser neurons exploit direct bandgap properties and strong nonlinear effects to enable ultrafast and efficient nonlinear response. Implementations include two-section gain saturable absorber excitable lasers,[87] semiconductor ring lasers,[88] quantum-dot lasers (QDLs)[53] and the Fabry–Perot lasers with saturable absorbers (FP-SA, Figure 3f).[52,89] QDLs and the FP-SA can respond to ps-level perturbation spikes and lead to fJ-level energy efficiency per spike,[52,53] outperforming most silicon photonic neurons. III-V laser neurons are also capable of emulating various biological neural behavior patterns. QDLs mimic biological neuron traits like excitatory/inhibitory response and refractory periods, while the FP-SA neurons achieve temporal signal integration and cascadable spiking for photonic SNNs.[52,53] Despite these merits, III-V laser neurons face critical barriers: excessive footprint (≈10^5 μm^2, 100 × larger than Si/Si_3N_4 neurons) and CMOS incompatibility (e.g. lack of foundry-standard epitaxial growth on Si), constraining scalable integration (**Table 1**).

4. Integrated Photonic Neural Networks

Given the extensive research in integrated PNNs, this part is structured into five sections to systematically categorize PNN architectures and emerging trends. Beginning with CNNs, the focus lies on optical implementations of convolutional kernels for parallel spatial feature extraction. This is followed by photonic MLPs, which omit convolution operations entirely, dedicating optical chips exclusively to implementing fully connected layers. The third section addresses diffractive neural networks (DNNs), emphasizing their evolution from free-space optics to compact on-chip implementations. Next, alternative architectures—including SNNs, Ising machines, and RC—are explored for their

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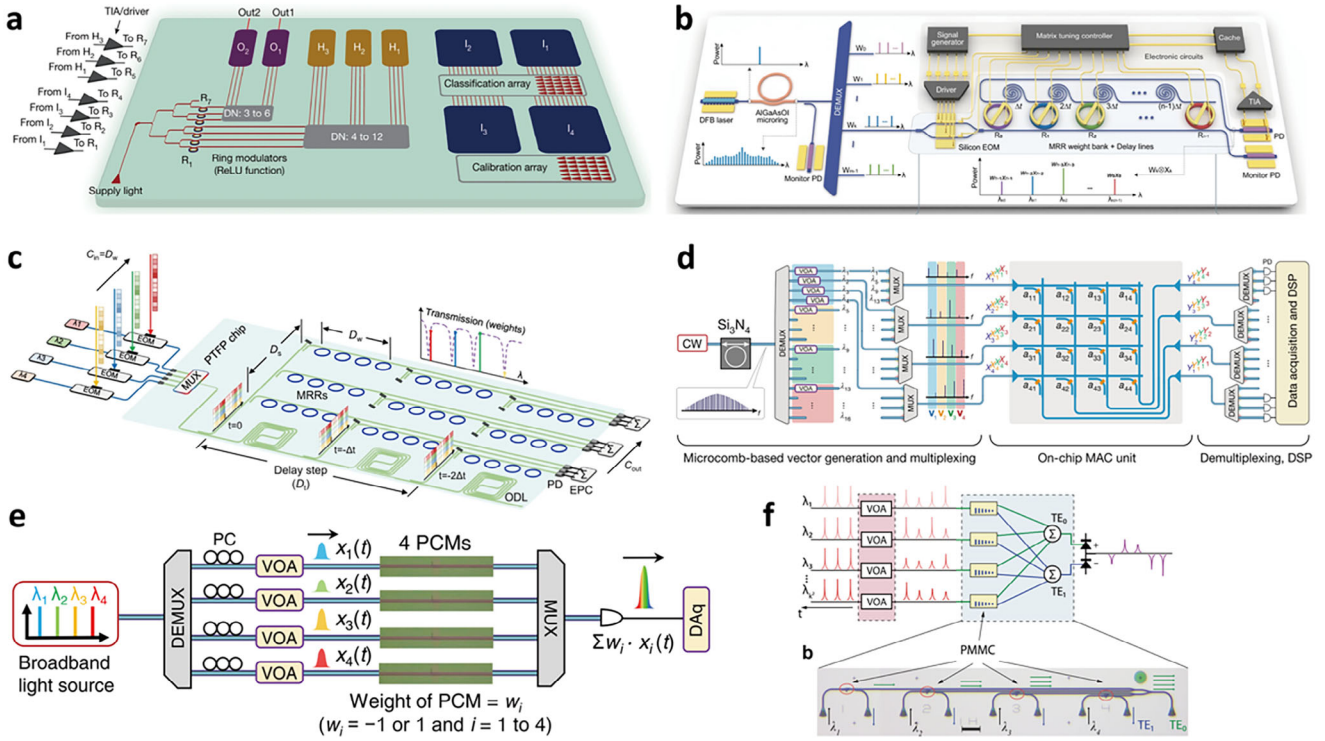


Figure 4. Integrated photonic CNN implementations. a), Three-layer optoelectronic deep neural network.[41] Linear computation is performed optically by EAMs and non-linear activation function is realized opto-electrically by MRRs. b), Integrated parallel photonic processing unit based on optical frequency combs,[105] implementing time-wavelength-interleaving convolution through MRR weight banks. c), Time-space-wavelength interleaving hybrid photonic tensor for high-dimensional data processing.[98] d), PCM memory arrays (waveguide crossings with PCMs on top), and integrated optical frequency combs-based photonic tensor core.[49] e), An array of four electrical regulated PCM cells on waveguides has been utilized to perform optical CNN.[99] f), Phase-change metasurface mode converter array to implement matrix multiplication for convolution.[106] a) Reproduced with permission.[41] Copyright 2022, Springer Nature. b) Reproduced with permission.[102] Copyright 2023, Springer Nature. c) Reproducedwith permission.[98] Copyright 2022, Springer Nature. d) Reproduced with permission.[49] Copyright 2021, Springer Nature. e) Reproduced with perimssion.[99] Copyright 2022, Springer Nature. f) Reproduced with permission.[106] Copyright 2021, Springer Nature.

unconventional computational paradigms and niche advantages. Finally, the fifth section delves into training strategies, highlighting optimization methods tailored for photonic hardware to meet the demands of large-scale AI models.

4.1. Photonic Convolutional Neural Network

CNNs rely on convolutional layers, where trainable kernels (filters) perform localized feature extraction. These kernels—small matrices—slide across input data (e.g. images) with fixed stride intervals, computing element-wise multiplications and spatial cumulations.[93] This generates feature maps encoding localized patterns (edges, textures, distinct patterns, etc.) critical for hierarchical processing. CNNs remain dominant in image recognition and feature extraction due to their translation-invariant, spatially aware architecture.[94] Convolutional layers are not intrinsically high-dimensional in terms of parameter count; rather, their computational burden stems from the large spatial dimensions of feature maps and the repeated sliding-window operations. To alleviate this, convolution is commonly reformulated into matrix multiplication via matrixization techniques (e.g., *im2col*[95]), where input patches are unfolded into column vectors and convolutional kernels are flattened into row vectors. The resulting matrices are then processed by optical devices that inherently support multiply-accumulate (MAC) operations. At the hardware level, strategies such as optical delay lines (ODLs) and wavelength-division multiplexing (WDM), i.e., time–wavelength interleaving,[96,97] have been proposed to implement sliding-window patching directly in the optical domain. In this scheme, optical signals at different wavelengths are staggered by delay lines and subsequently superimposed in their overlapping regions. Additional measures, such as space-division multiplexing, will be discussed in the following section.

In PICs, convolutional layers are mapped into matrix multiplication by leveraging optical devices' inherent capability to perform multiply-accumulate (MAC) operations.[41,98,99] The architecture in **Figure 4a** adopts a space-division multiplexing scheme for optical convolution: input images are partitioned into four overlapping sub-images, each processed by dedicated first-layer neurons. This approach enabled a monolithic three-layer PIC implementation achieving sub-nanosecond end-to-end image classification. In 2021, Xu et al. reported a time-wavelength-interleaving convolutional architecture using off-chip discrete optical components (optical frequency combs[100,101] and electro-optic modulators), achieving a computational performance of

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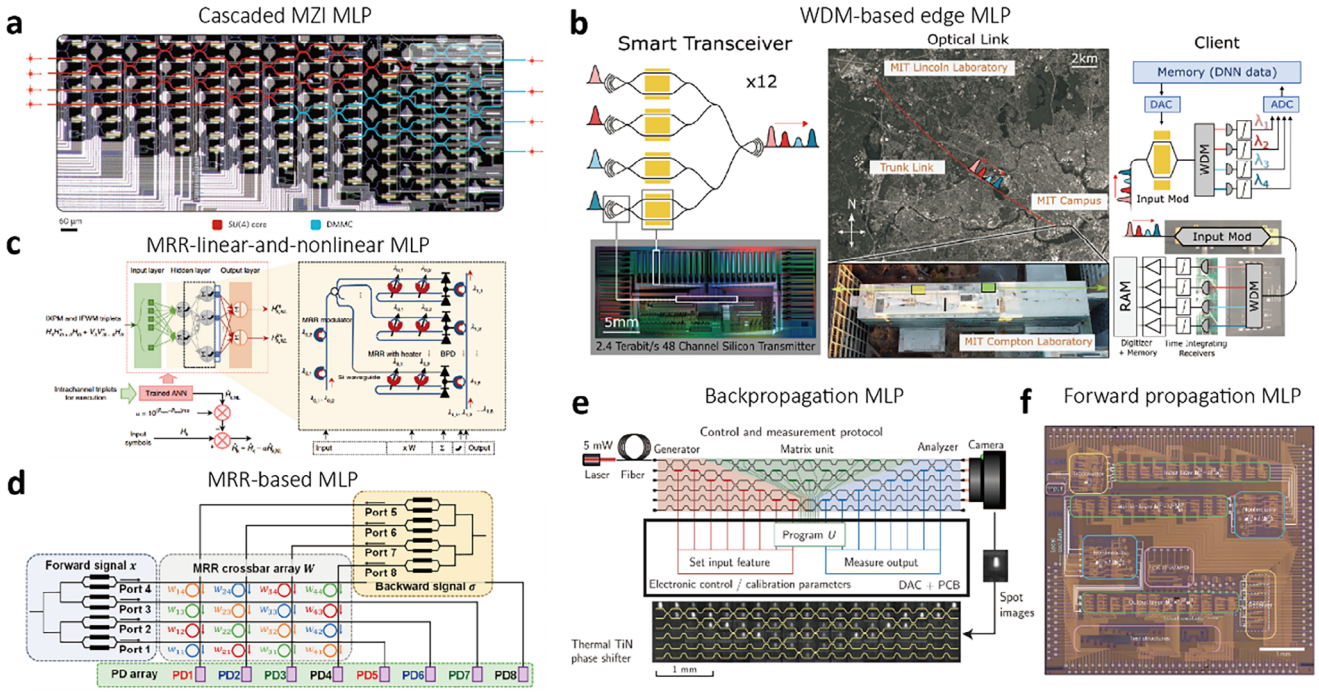


Figure 5. Integrated photonic MLP implementations. a), The PIC integrates 56 MZIs,[38] where red-highlighted MZIs perform matrix multiplication and blue-highlighted MZIs implement amplitude attenuation. b), An optical fiber over 86 km was utilized to transmit weighted data from cloud-based "smart transceivers" to the client.[107] c), MRR-based hidden layer, with MRR implementing linear and nonlinear operations, respectively.[65] d), 4 × 4 MRR crossbar-based PNN for matrix-vector multiplication (MVM).[108] e), An experimentally trained three-layer, four-port PNN.[43] in situ backpropagation is achieved by utilizing programmable phase shifters and off-chip power monitors. f), A fully integrated 3-layer PNN with linear and non-linear implemented in a single chip.[109] a) Reproduced with permission.[38] Copyright 2017, Springer Nature. b) Reproduced with permission.[107] Copyright 2022, AAAS. c) Reproduced with permission.[65] Copyright 2021, Springer Nature. d) Reproduced with permission.[108] Copyright 2022, American Chemical Society. e) Reproduced with permission.[43] Copyright 2023, AAAS. f) Reproduced with permission.[109] Copyright 2024, Springer Nature.

up to 11 TOPS.[102] Figure 4b demonstrates an integrated time-wavelength-interleaving convolution, with on-chip kernels realized through MRR weight banks and optical delay lines. Other studies have adopted delay lines to leverage analogous time-interleaving strategies, enabling parallel photonic convolution kernels through temporal multiplexing.[103,104] Figure 4c indicates a spatial-wavelength-delay hybrid architecture for direct high-dimensional data processing via optical-domain tensor operations. Figure 4d leverages spatial-wavelength multiplexing, encoding input data across an optical frequency comb's wavelengths. By integrating PCMs for non-volatile weighting, this design achieves a monolithic photonic accelerator operating at Tera-MAC/s (10^{12} multiply-accumulate operations per second). Figure 4e illustrates another PCM-based CNN using electrical regulation of PCMs' states (contrasting with the optical switching method in Figure 4d), achieving 4-bit weight precision through controlled phase transitions of PCMs. In Figure 4f, a PCM metasurface leverages four multimode converters to enable 64-level modal contrast resolution, showcasing enhanced analog programmability of PCMs for optical neural processing.

4.2. Photonic Multilayer Perceptron

This section focuses on non-convolutional optical neural networks implemented via PICs using MZIs and other photonic devices. In contrast to convolutional approaches, these architectures dedicate photonic hardware exclusively to fully connected (FC) layers, bypassing transformations like matrix-mapped convolutions. FC layers—foundational building blocks of neural networks—establish dense interconnections between all neurons in adjacent layers, enabling the network to model intricate global dependencies within data. By leveraging photonics for parallelized matrix operations, such designs prioritize scalability and linear algebraic efficiency while retaining universal approximation capabilities.

Figure 5a represents a seminal advance in optical neural networks, introducing an MZI-based all-optical architecture that established a blueprint for scalable photonic deep learning. Figure 5b implements an edge-optimized photonic architecture, integrating optical transceivers, optical links, and client-side processing for end-to-end photonic inference at 40 aJ/MAC—a benchmark for energy-efficient edge-AI photonics. An alternative photonic MLP design leveraging on-chip MRRs for linear and nonlinear operations, enabling full complex-valued functionality, as depicted in Figure 5c. Figure 5d–f showcases advances in on-chip training methodologies (detailed in Section 4.5) of photonic MLPs. Figure 5d achieves on-chip backpropagation by exploiting the native transpose operation of MRR crossbar arrays—enabling a straightforward on-chip training approach. Figure 5e,f utilizes MZI-based architectures akin to Figure 5a but diverge critically in training methodology. Specifically, Figure 5e implements

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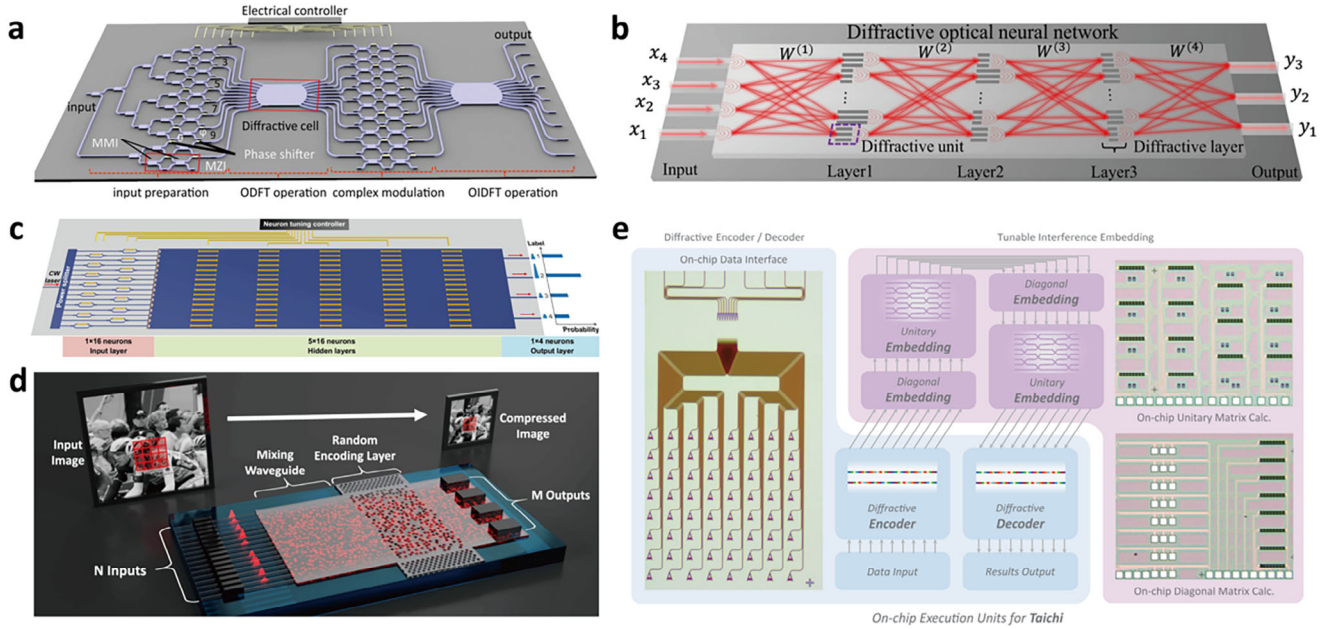


Figure 6. Integrated photonic DNN implementations. a), An integrated DNN[113] includes four parts: input preparation, optical discrete Fourier transform (ODFT) operation, modulating amplitude/phase, and optical inverse discrete Fourier transform (OIDFT). b), An implementation of on-chip DNN[42] with each diffractive unit (a single neuron) comprising a slot group consisting of three identical silicon slots filled with silicon dioxide. c), A programmable DNN chip[44] with input, hidden and output layers. The input layer consists of 16 intensity modulation units for loading data, while the hidden layer comprises 5 layers, each containing 16 tunable diffraction units. The output layer features 4 output ports. d), A photonic image encoder[114] compresses images by having N inputs enter a multimode waveguide region, where disordered scattering occurs in a random encoding layer. This process results in M outputs ($M < N$). e), The Taiji chip[45] consists of two main components: diffraction units for encoding and decoding high-dimensional input data, and a tunable MZI array for feature embedding. a) Reproduced with permission.[113] Copyright 2022, Springer Nature. b) Reproduced with permission.[42] Copyright 2023, Springer Nature. c) Reproduced with permission.[44] Copyright 2024, Springer Nature. d) Reproduced with permission.[114] Copyright 2024, Springer Nature. e) Reproduced with permission.[45] Copyright 2024, AAAS.

layer-wise backpropagation, constrained to sequential single-layer computation and reliant on off-chip infrared camera readout, necessitating three operations per iteration. In contrast, Figure 5f demonstrates forward-propagation-based training, featuring a monolithic three-layer network with integrated photodetectors and optoelectronic nonlinearities. This end-to-end photonic design slashes iterations to 2 operations/cycle by eliminating off-chip data transfer. In addition, the recently debuted quad-core photonic processor[60] pioneers optoelectronic co-designed neuromorphic systems, executing convolutional networks and multilayer perceptrons with FP-32-equivalent accuracy in benchmark tests against digital platforms. This breakthrough underscores photonics' viability as a high-precision, energy-efficient alternative to electronic AI accelerators, establishing a scalable framework for next-generation photonic AI hardware.

4.3. Photonics Diffractive Neural Network

Free-space optical computing, which utilizes phase modulation of light in space, emerged as a disruptive paradigm. This approach underpins DNNs that process data through cascaded diffractive layers performing linear transformations with unprecedented spatial parallelization.[110,111] However, subwavelength structural requirements impose THz-range wavelength modulation and bulky footprints. Two strategies address this: free space metasurface-based DNNs miniaturize modulation units, enabling shorter-wavelength operation and tightly stacked layers—though at increased fabrication complexity; and integrated photonic processors implement diffractive networks on-chip through programmable wavefront shaping, trading spatial degrees of freedom for CMOS-compatible scalability.[45,112] Unlike free space metasurfaces that rely on low-speed (Hz–kHz) modulators such as Spatial Light Modulator(SML) and Digital Micromirror Device(DMD) for data loading, on-chip diffractive neural networks can leverage a variety of ultra-high-speed modulation mechanisms available in silicon photonic platforms—such as thermo-optic modulators (≈kHz), carrier dispersion modulators (≈MHz–GHz), and electro-optic modulators (GHz)—to achieve extremely high-speed data loading. This capability is functionally analogous to data loading techniques used in on-chip diffractive photonic chips, thereby enabling higher computational throughput and greater computational density. **Figure 6** demonstrates on-chip DNNs leveraging integrated photonics for compact, energy-efficient optical computing. Figure 6a presents a radically optimized architecture using two multimode waveguide-based diffractive units and MZIs—slashing the traditional N^2 MZI count to N. This innovation reduces both energy use and footprint by an order of magnitude. Figure 6b implements a static three-layer DNN using fixed SiO_2-filled etched silicon nanosctructures, where neuron configurations are permanently locked post-fabrication. In contrast, Figure 6c introduces a seven-layer reconfigurable variant (1 input, 5 trainable hidden layers, 1 output) with tunable diffractive units, enabling

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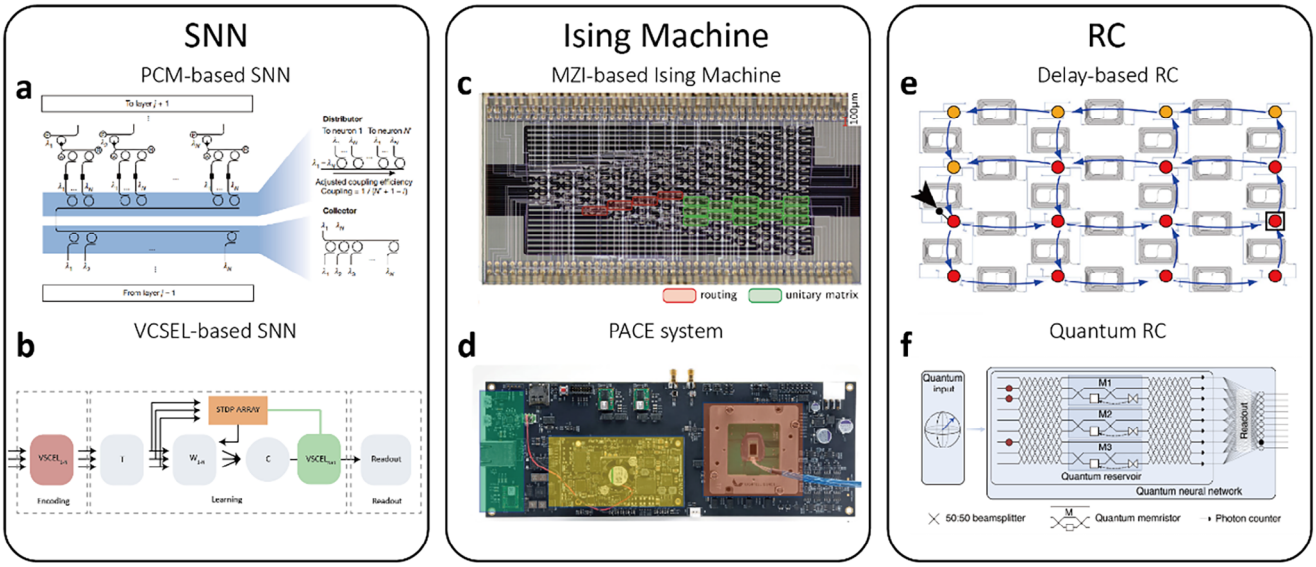


Figure 7. Implementations of integrated photonic SNN, Ising machine, and RC. a), A PCM-based SNN[48] using wavelength-division multiplexing (WDM) architecture. The collector combines optical pulses from the previous layer, while the distributor assigns the input signal equally to PCM synapses of the next layer. b), A SNN[125] based on vertical-cavity surface-emitting lasers (VCSELs) and vertical-cavity semiconductor optical amplifiers (VCSOAs). The STDP array is realized by VCSOAs. c), An integrated nanophotonic coherent Ising machine[56] based on a programmable nanophotonic processor (PNP). The PNP consists of 88 MZIs and 176 independently controlled phase shifters (PSs). The entire system receives optical signals from the PNP through a photodiode array, which then calculates and updates the phase of each PS in the PNP for the next iteration via a feedback loop. d), The PACE system.[59] This system is a large-scale integrated photonic accelerator composed of over 16 000 photonic elements. It employs advanced 2.5D optoelectronic heterogeneous integration to seamlessly combine photonic and electronic chips. e), An integrated passive photonic RC system[54] based on delay lines and multimode interferometers (MMIs). f), A quantum RC[55] based on quantum memristors, with a feedback loop that controls the phase in one of the arms of an MZI. a) Reproduced with permission.[48] Copyright 2019, Springer Nature. c) Reproduced with permission.[56] Copyright 2020, Optica Publishing Group. d) Reproduced with permission.[59] Copyright 2025, Springer Nature. e) Reproduced with permission.[54] Copyright 2014, Springer Nature. f) Reproduced with permission.[55] Copyright 2022, Springer Nature.

dynamic weight adjustment and in situ training. Each diffractive unit is rendered tunable by locally adjusting the effective refractive index in the corresponding region of the hidden layers, achieved through microheaters that precisely control the temperature near each unit. During training, an FPGA co-processor dynamically adjusts on-chip diffractive units via optoelectronic feedback, enabling in situ weight updating for adaptive photonic learning. Figure 6d,e illustrates the use of diffractive units for encoding/decoding input data. In Figure 6d, silicon-etched holes enable a low-power image compression solution, achieving high-speed compression with power consumption several orders of magnitude lower than electronic devices. A recently proposed Taichi photonic chiplet, a breakthrough in scalable optical computing, is shown in Figure 6e. By integrating engineered silicon slots for hybrid diffractive-interference operations, Taichi combines ultra-dense neuron integration (millions per chip) with unprecedented power efficiency (160 TOS/W). Its architecture leverages the parallel nature of light to perform encoding, computation, and decoding on-chip, bypassing electronic data movement penalties.

4.4. Photonic Spiking Neural Network, Ising Machines, and Reservoir Computing

Beyond conventional neural architectures, photonic computing extends its versatility to specialized computational models critical for next-generation AI, such as SNNs, Ising Machines, and RC, which will be reviewed in this section. As a third-generation neural network paradigm, SNNs synergize the structural complexity of ANNs with the biologically inspired dynamics of neuronal spiking, unlocking transformative potential for energy-efficient adaptive control systems[115] and temporal signal processing.[116] Mimicking the brain's synaptic architecture,[117] SNNs encode information in event-driven spikes—presynaptic stimuli modulated by synaptic weights (e.g., via spike-timing-dependent plasticity, STDP[57]) and integrated postsynaptically in dendritic structures. Only when spatiotemporal input exceeds a threshold does a neuron emit a spike (integrate-and-fire (IF)[52] or leaky integrate-and-fire (LIF)[118]), a sparse activation mechanism that slashes energy consumption by maintaining quiescence during idle periods. Photonic implementations leverage devices like PCMs for nonvolatile synaptic weighting[47,55] and vertical-cavity semiconductor optical amplifiers (VCSOAs)[115] for ultrafast spike generation. **Figure 7a** demonstrates a fully integrated all-optical SNN using wavelength-division multiplexing (WDM) to scale network complexity, enabling both supervised and unsupervised learning through reconfigurable PCM devices. Figure 7b presents an on-chip STDP realization, deploying VCSOAs to optically emulate synaptic plasticity.

Ising model,[119] initially a framework for understanding ferromagnetic phase transitions, now serves as a cornerstone for solving NP-hard combinatorial optimization problems—tasks intractable for classical von Neumann architectures. While

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Table 2. Comparison of performance metrics for various integrated photonic neural networks in the experiments.

Network Structure	Hardware architecture	Compute power [TOPS]	Area [mm^2]	Energy efficiency [TOPS/W]	Compute density [TOPS/mm^2]	Speed [GHz]	Platform	Programmable	Matrix scale	On-chip training	On-chip nonlinearity
CNN[102]	fiber	11.3	–	1.25	–	62.9	fiber	Yes	9 × 10	No	No
CNN[49]	Tenser-Core	2	≈4	infintiy	0.5	2	SOI	Yes	4 × 4	No	No
CNN[105]	MRRs	0.136	0.131	0.00152	1.04	17	SOI	Yes	1 × 4	No	No
CNN[127]	Nanobeam array	4.8	–	1.32	–	–	SOI	Yes	3 × 3	No	No
MLP[41]	on-chip VOA	22	9.3	2.9	0.27	22	SOI	Yes	5 × 6	No	No
MLP[65]	InP-SOA	0.64	–	0.24	–	10	InP	Yes	8 × 8	No	No
MLP[38]	MZI-mesh	0.001	1.098	–	0.00091	100	SOI	Yes	4 × 4	No	No
MLP[109]	MZI-mesh + ring	0.53	33	0.1	0.016	–	SOI	Yes	6 × 6	Yes	Yes
MLP[126]	Tenser-Core	180	18	3.33	10	–	SOI	Yes	3 × 3	No	No
DNN[45]	diffraction + MZI	50380	28.66	160	1757	2	SOI	Partially	64 × 54	No	No
DNN[44]	Diffraction	217.6	0.49	7.28	447.7	100	SOI	Yes	16 × 16	Yes	No
DNN[42]	diffraction + PS	1380	0.3	90.9	4600	100	SOI	Partially	–	Yes	No
DNN[113]	diffraction + MZI	30	0.53	–	56.6	–	SOI	Partially	10 × 10	No	No
SNN[48]	Ring + PCM	0.0003	2.72	0.035	0.00011	0.005	SiN	Yes	15 × 4	Yes	Yes
RNN[54]	delay line	–	–	–	–	–	SOI	Yes	–	No	No
Ising[56]	MZI-mesh	–	1.098	–	–	1	SOI	Yes	4 × 4	No	No

MRR, microring resonator; SOA, semiconductor optical amplifier; MZI, Mach–Zehnder interferometer; PS, phase shifter; PCM, phase-change material; TOPS, tera operations per second; SOI, silicon on insulator.

low-dimensional Ising systems yield to analytical solutions, higher-dimensional variants (3D+) demand novel computing paradigms. Photonic Ising machines[120,121] address this by emulating natural energy minimization: these systems iteratively evolve toward the Ising Hamiltonian's ground state, effectively solving optimization challenges like Max-Cut or prime factorization. Figure 7c exemplifies this with an on-chip coherent Ising machine, where arrays of MZIs encode arbitrary Ising graphs through unitary matrix decomposition. Operating at GHz clock rates, this architecture harnesses light's parallelism to sample energy landscapes orders of magnitude faster than electronic solvers. Building on this, Figure 7d introduces PACE (Photonic Accelerated Computing Engine), an optoelectronic hybrid system integrating optical MAC units. By synergizing MZI-based photonics with electronics, PACE achieves >100 × reductions in latency and compute time versus GPUs, enabling real-time solutions to complex Ising instances. This advancement is expected to facilitate the exploration of unique computational paradigms and system architectures leveraging optoelectronic computing systems.

RC circumvents the training complexity of traditional ANNs by replacing densely connected layers with a fixed high-dimensional reservoir,[122–124] drastically reducing trainable parameters to only the output layer. This architecture—comprising an input layer, static reservoir, and adaptable output—enables rapid training while retaining nonlinear expressive power for chaotic time-series prediction and speech recognition. Photonic implementations exploit optical nonlinearities and parallelism to realize efficient RC systems. Figure 7e demonstrates the first fully passive integrated photonic RC, using 16 silicon nodes interconnected by 2 cm delay lines and MMIs to create recurrent feedback loops, leveraging light's propagation dynamics as a computational resource. Figure 7f pioneers a quantum-enhanced RC, where input data is scrambled via fixed beam splitters and processed by three photonic quantum memristors—nonlinear optical elements that endow the system with both nonlinearity and short-term memory. The memristors' outputs are re-scrambled and measured by a photon counter, enabling low-power, high-speed temporal processing. These photonic RCs harness passive optical components and quantum-optical effects to outperform electronic counterparts in latency and energy efficiency, offering a scalable pathway for real-time edge computing and adaptive signal processing in resource-constrained environments (**Table 2**).

4.5. Photonic Training Implementations

While photonic chips remain constrained in general-purpose computing relative to their electronic counterparts, recent efforts have shifted from prioritizing specialized acceleration of ANN inference workloads to enabling photonic-native training—a paradigm demanding tight hardware-algorithm co-design. Training PNNs hinges on real-time gradient computation, typically necessitating optical architectures capable of backpropagating errors.

As illustrated in **Figure 8a**, Cheng et al. exemplify a pragmatic hybrid strategy for PNN training, balancing on-chip optical advantages with off-chip electronic flexibility. Their five-layer tunable diffractive network executes forward propagation optically, while delegating gradient calculation and weight updates to an external electronic computer. This approach circumvents the critical challenge of in situ photonic gradient computation—limited by the absence of native optical differentiation modules and dynamic nonlinearities—while still incorporating photonic

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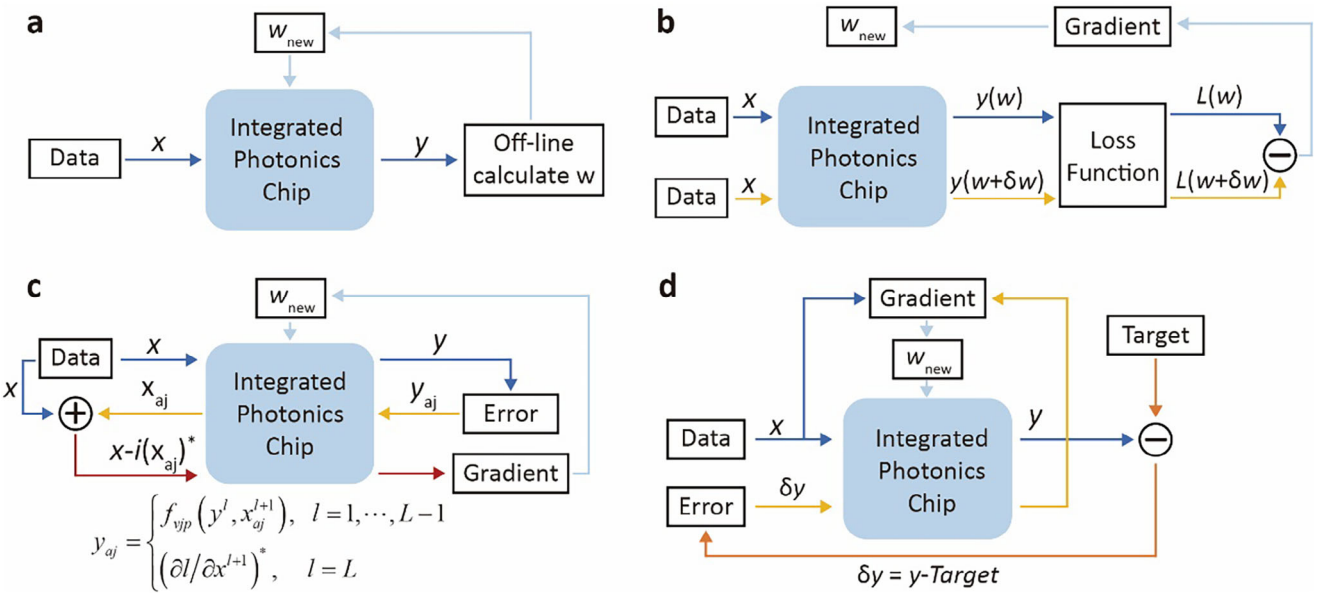


Figure 8. PNN training implementations. a) The photonic chip processes the input data *x* to generate the output *y*, the chip is treated as a 'black box' with numerous trainable weight parameters. The weights are calculated on an FPGA, and then the on-chip weights are updated.[44] b) The input data *x* is first fed forward into the photonic chip, and the weights are then perturbed. The same input *x* is fed forward again, and the difference between the two outputs is used to compute the weight gradient, which is subsequently used to update the weights.[109] c) The input data *x* is first fed forward into the chip, producing the output *y*. The corresponding adjoint matrix y_{aj} is then computed and fed backward into the chip to obtain x_{aj}. The summation of *x* and x_{aj} is then reintroduced forward into the chip to compute the gradient and update the weights.[44] d) The input data *x* is first fed into the chip, and the output *y* is compared with the target to compute the error (δy). Since the chip operates as a spatially symmetric system, forward propagation can replace backpropagation. The error result is then reintroduced forward into the chip, and the obtained output (δx) is combined with the input *x* to calculate the gradient for weight updating.[131]

hardware imperfections (e.g., fabrication variances, thermal crosstalk) into the training loop. By co-optimizing the physical chip's behavior within the digital backpropagation framework, the hybrid system achieves superior inference accuracy compared to purely simulation-based training, which often overlooks real-world photonic non-idealities. Bandyopadhyay et al. (Figure 8b) pioneered fully on-chip PNN training via a stochastic gradient descent (SGD) framework,[109] bypassing the need for external electronic gradient computation. Their three-layer fully connected network executes end-to-end optical training by calculating gradients for individual weights per iteration—leveraging direct photonic measurements of loss perturbations. Unlike backpropagation, which requires simultaneous gradient updates across all parameters, SGD's sequential approach aligns with photonic hardware's inherent parallelism and analog nature, enabling the training process, including both linear and nonlinear operations, to reside entirely on-chip. However, this method trades convergence speed for hardware compatibility: the stochastic sampling of gradients increases iteration counts compared to batch gradient descent. Yet, photonics' sub-nanosecond latency mitigates this tradeoff, as the absence of RC delay.[128,129] By eliminating electronic bottlenecks in weight updates, this work marks a critical step toward self-contained photonic learning systems. Additionally, Pai et al. (Figure 8c) achieved 94% MNIST accuracy by re-engineering backpropagation for optical hardware.[130] Their system computes gradients through dual forward passes: a primary inference pass and a secondary error-propagation pass, with an infrared camera monitoring photonic outputs to derive weight updates without electronic simulation. While successful, this method's reliance on offline camera readout introduces latency, limiting real-time adaptability. Addressing this, Xue et al. (Figure 8d) devised a symmetry-enabled gradient architecture, where input data (*X*) and error signals (δ) propagate concurrently through mirrored optical paths.[131] By exploiting reciprocal light transmission in photonic circuits, gradients emerge inherently from forward-only propagation, bypassing traditional backpropagation's bidirectional computational overhead. This approach not only simplifies experimental implementation but also reduces accumulated errors from cascaded nonlinearities, achieving comparable accuracy with lower optical noise.

5. Emerging Ideas and Outlooks

Photonic neural chips are revolutionizing AI acceleration across the computing hierarchy, tailored to meet the divergent demands of cloud data centers and edge client systems.[132] For cloud infrastructure, where exascale throughput and dynamic reconfigurability are paramount, photonic systems leverage wafer-level integration (**Figure 9**) heavily reliant on WDM technology to maximize data density and throughput. To fully exploit WDM's capacity, advanced foundry processes enable multi-layer Si-Si_3N_4 waveguides: ultra-low-loss Si_3N_4 facilitates centimeter-scale optical routing, while doped silicon supports GHz-speed electro-optic modulation and detection. Time-delay network multiplexing further amplifies logical network scale by reusing physical waveguides with staggered optical pulses, decoupling computational capacity from chip footprint. These innovations sustain

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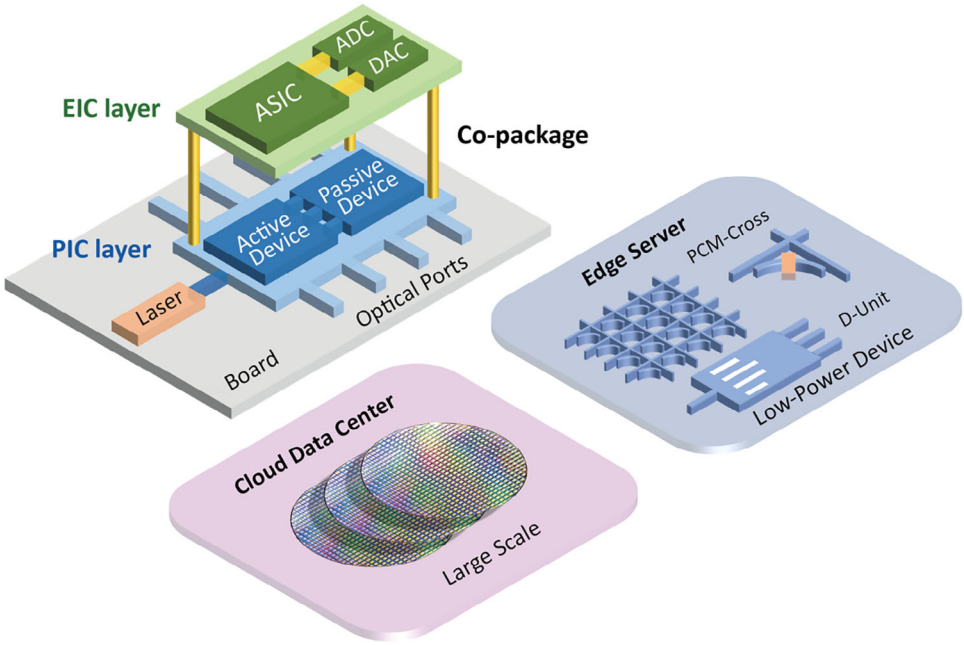


Figure 9. Schematic of the future photonic computing system. The system consists of an external laser, a photonic integrated circuit (PIC), and an electronic integrated circuit (EIC). The external laser is coupled to the PIC via couplers. The PIC containing active and passive components, and the EIC, which includes application-specific integrated circuits (ASICs) as well as AD/Das and transimpedance amplifiers (TIAs), are co-packaged on the same substrate. The ASICs are used to process information and control the AD/ DAs. The PIC features optical ports that facilitate interconnections with other PICs. Different application scenarios, such as cloud data centers and edge servers, impose distinct requirements on PICs. Data centers demand a sufficiently large-scale PIC, potentially achieved through wafer-scale and chiplet photonic integration, whereas edge applications prioritize low power consumption, which can be accomplished using PCMs and diffractive units (D-Units) in DNN.

distributed training of trillion-parameter models for hyperscale tasks like real-time recognition analytics.

At the edge, photonic hardware prioritizes ultra-low power consumption and compact computational density for offline, battery-operated applications like autonomous driving and IoT. Here, passive PCMs[133] (Figure 9) provide nonvolatile, reprogrammable synaptic functionality[67,134] with femtojoule-per-switch energy efficiency, eliminating static power draw. Paired with on-chip diffractive units (D-Units)—harnessing subwavelength light-matter interactions for passive, parallel, high-speed matrix operations—edge systems achieve TOPS/mm^2 performance with limited active components. While edge devices demand selective reconfigurability for fine-tuning of models and firmware updates, cloud-edge synergy ensures both domains meet sub-microsecond latency thresholds for real-time responsiveness.

Achieving training stability in PNNs—critical for both cloud and edge deployments—demands co-optimization of device physics, environmental control, and algorithmic resilience. Temperature-sensitive photonic components like MRRs and PSs necessitate active thermal management, often addressed via thermoelectric coolers (TECs) to stabilize operating conditions. Beyond brute-force cooling, self-calibration techniques dynamically counteract thermal crosstalk by tuning PSs in real time, leveraging closed-loop feedback to maintain weight integrity.[135] For precision, hardware-software co-design strategies are emerging: 1) Optical link monitoring tracks and compensates for weight drift caused by environmental perturbations and thermal crosstalk, using integrated photodetectors to stabilize analog computations;[62] 2) Weight pruning reduces parameter sensitivity to noise, enabling robust inference without TEC overhead.[136] Beyond improving inference reliability, pruning substantially decreases the number of active photonic components, memory usage, and control complexity, collectively boosting power and area efficiency.[137] However, aggressive pruning risks overly constraining the model's parameter space, potentially degrading accuracy. Structured techniques like block-circulant pruning,[138] exemplify this trade-off: while larger block sizes enable higher compression ratios, they also increase the likelihood of performance loss. To navigate this balance, hardware-aware retraining and adaptive pruning strategies become critical, ensuring efficiency gains are achieved without sacrificing model fidelity. Architectural innovations like RC further sidestep precision limitations by minimizing trainable parameters—only the readout layer is tuned, inherently insulating training outcomes from analog non-idealities. Together, these advances—spanning thermal engineering, adaptive calibration, and neuromorphic architectures—propel photonic networks toward deployable reliability. Subsequent sections will explore how device-level and system-level innovations can amplify these gains, scaling photonic neural networks for real-world robustness.

5.1. On-Chip Lasers

The quest for on-chip lasers—a linchpin for fully integrated photonics circuits—remains a pivotal challenge in silicon photonics. Despite their dominance in foundry processes, silicon (Si)

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platforms exclude integrated lasers due to Si's indirect bandgap,[139] which stifles optical gain and efficient light emission. Current solutions rely on externally coupled lasers via optical fibers, but this introduces bulky packaging, alignment challenges, and scalability limits. To overcome silicon's inherent constraints, researchers pivot to III–V semiconductors (e.g., GaAs, InP), whose direct bandgaps enable high-efficiency lasing.[140] However, III-V materials suffer from waveguide losses and low refractive index contrast, making standalone III-V photonic integration impractical.[141] Hybrid approaches merge III-V gain media with silicon's low-loss waveguides through two strategies: 1) Heterogeneous integration: bonding III-V wafers to silicon-on-insulator (SOI) substrates, enabling high-performance lasers with CMOS compatibility;[142–145] 2) Monolithic integration: direct epitaxial growth of III-V layers on Si, bypassing bonding but grappling with lattice-mismatch-induced defects.[146–149]

Parallel efforts explore Group IV materials like germanium (Ge) and germanium-tin (GeSn) alloys. Though CMOS-compatible, pure Si and Ge require bandgap engineering to rival III-V efficiency, including subwavelength structures of Si,[150,151] and N-type doping of Ge[152,153] and GeSn alloying.[154,155] Yet, these Group IV lasers lag behind III-V counterparts in threshold current density and emission efficiency, necessitating breakthroughs in material synthesis and cavity design. As photonic foundries gradually embrace hybrid platforms, the integration of efficient on-chip lasers will catalyze breakthroughs in optical AI, quantum interconnects, and ultra-low-latency telecom.

5.2. Integrated Optical Amplifiers

Insertion loss in deep PNNs—especially across multi-layer interconnects—poses a critical barrier to scalability. While boosting on-chip laser power offers one remedy, integrated optical amplification emerges as a transformative alternative. Among on-chip solutions, semiconductor optical amplifiers (SOAs) stand out, leveraging population inversion in III-V materials (e.g., InP, GaAs) to amplify light via electrical pumping.[156,157] Li et al. demonstrated a two-stage amplified large optical cavity, reducing optical field confinement to achieve 419 mW output power and 26.23 dB gain at 1 mW input—a milestone in high-power photonic integration.[158] Eyal et al. advanced this further with quantum dot SOAs (QD-SOAs), delivering 22 dB static gain while maintaining signal integrity for 28 Gbit s^{-1} data streams, crucial for high-speed optical interconnects.[159] Despite their ease of integration (akin to on-chip lasers) and electrical pumping efficiency, SOAs face tradeoffs: high amplified spontaneous emission (ASE) noise and saturation-limited gain/power restrict their use in low-noise, high-power applications.

On the other hand, rare-earth-ion-doped waveguide amplifiers present a low-noise alternative to SOAs, leveraging erbium (Er) or other rare-earth ions to enable population inversion via optical pumping.[160] By embedding these ions within waveguide materials, researchers achieve broadband amplification with minimal noise—critical for photonic networks requiring high signal fidelity. Mu et al. pioneered this approach by coating a silicon nitride (SiN) waveguide with an erbium-doped alumina gain layer, realizing 18 dB gain across a 70 nm bandwidth, compatible with diverse passive photonic platforms.[161] Liu et al. further advanced this approach via directly doping erbium ions into SiN waveguides to deliver 30 dB gain and 145 mW output power, rivaling commercial erbium-doped fiber amplifiers (EDFAs) in performance.[162] While rare-earth-ion-doped amplifiers excel in noise suppression and moderate power scalability, their reliance on external optical pumping complicates on-chip integration compared to electrically driven SOAs. Additionally, challenges in ion uniformity and pump-light coupling efficiency hinder mass production readiness.

5.3. Modulators

Silicon photonic modulators in mainstream foundry process design kits (PDKs) predominantly exploit the plasma dispersion effect (PDE), altering carrier concentrations to modulate phase or amplitude at speeds exceeding 100 Gb s^{-1}.[163–166] While PDE-based devices excel in modulation bandwidth, their millimeter-scale footprints impose impractical area overheads for large-scale photonic networks. To transcend this limitation, researchers are integrating novel electro-optic materials with silicon photonics, such as silicon-germanium alloys (SiGe),[167–170] III–V semiconductors,[171–175] 2D materials,[176–180] and ferroelectric materials.[181–185] These materials offer opportunities for enhancing the performance of silicon photonics modulators.

SiGe modulators, leveraging the Franz–Keldysh effect (FKE) or quantum-confined Stark effect (QCSE), achieve electroabsorption modulation in sub-10 μm^2 footprints with low power consumption. However, their narrow optical bandwidth restricts multi-wavelength applicability. III-V semiconductors, such as InP and GaAs modulators, capitalize on strong inherent electro-optic effects for high efficiency, yet struggle with high-speed operations. 2D materials, particularly graphene, whose ultra-high carrier mobility and broadband response promise THz-class modulation, but immature integration techniques hinder reproducible fabrication. Ferroelectric materials, including thin-film lithium niobate ($LiNbO_3$) and barium titanate ($BaTiO_3$) exhibit exceptional Pockels coefficients, enabling low-voltage, high-speed phase modulation. However, they face challenges in integration complexity and high propagation loss when integrated with silicon photonics.

While bulk ferroelectric modulators dominate commercial markets, their incompatibility with silicon photonics relegates them to discrete off-chip deployments. Future advancements demand material-device co-design—optimizing heterogeneous interfaces, loss mitigation, and driver circuitry—to bridge the gap between high-speed optical modulation and scalable silicon photonics platforms.

5.4. Photodetectors

On-chip photodetectors are pivotal for optical computing, demanding high speed, minimal dark current, and CMOS compatibility. Germanium (Ge) dominates mainstream silicon photonics due to its narrow bandgap (enhanced C/L-band absorption) and mature epitaxial growth. Foundries offer vertical Ge PIN photodetectors with speeds up to 50 GHz,[186] but their thick intrinsic

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regions (≈µm-scale) incur slow carrier diffusion, capping high-frequency performance. Transitioning to lateral PIN structures slims the intrinsic region to ≈100 nm, achieving record speeds of 265 GHz,[34] yet Ge still grapples with high dark current (≈nA to µA) and limited spectral range.

III-V semiconductors (e.g., InGaAs) emerge as superior alternatives, offering lower dark currents, broader wavelength detection, and dual functionality as lasers. Integrating III–V materials on silicon requires heterogeneous integration, through direct growth,[187] wafer bonding,[188] and transfer printing.[189] Due to the lattice mismatch between silicon and III-V materials, direct growth necessitates the pre-growth of a buffer layer before the III–V material can be deposited, allowing for the fabrication of III–V photodetectors with a bandwidth of 40 GHz.[187] Wafer bonding combines silicon wafers and III-V materials using van der Waals forces under suitable pressure and temperature conditions. Due to the high quality of interfaces, this method can be used to fabricate photodetectors with bandwidths exceeding 67 GHz.[190] Transfer printing uses a DVS-BSB layer to achieve device-level transfer of III–V photodetector. The lower temperature requirements in this process lead to improved device performance. Meanwhile, 2D materials like graphene revolutionize photodetection with ultra-wideband response (visible to THz) and passive integration. Metal-graphene-metal structures achieve 262 GHz bandwidth,[191] while plasmonic enhancements boost responsivity to 0.5 A W^{-1} at 110 GHz.[192] Advances in CVD wafer-scale growth[193] signal industrial viability, though yield and uniformity challenges persist. Recent advances have achieved heterogeneous integration of perovskite-based PDs onto SiN photonic chips,[194] overcoming critical barriers in merging disparate nanophotonic materials for multifunctional on-chip systems.

5.5. Passive Components

Passive Neuromorphic Devices: Passive photonic neuromorphic devices form the computational backbone of on-chip neuromorphic systems, enabling linear weighting and nonlinear activation through intrinsic material properties. PCMs and 2D materials dominate this space, offering sub-nanosecond optical modulation with (sub-)nanometer-scale thickness for ultra-dense integration. PCMs like GST excel in nonvolatile programmability, where reversible phase transitions enable in-memory photonic computing. However, repeated switching induces phase separation, degrading optical contrast. Material optimization (e.g., low-temperature doping[46] and pure element PCM[195]) and nanostructure design (e.g., nanodisks[196]) are effective and promising methods to address this challenge. 2D materials (e.g., graphene,[197] TMDCs[198]) provide ultrafast nonlinearity (ps to fs) and broadband response, yet their adoption hinges on wafer-scale monolayer synthesis and CMOS-compatible patterning.

On-chip Diffractive Units: By structuring diffractive layers to spatially modulate optical phases, DNNs execute MVMs through interference, exploiting wavelength-scale parallelism for tasks like image classification and optical encryption. Their ultra-low latency (speed-of-light operation) and zero static power consumption make them ideal for edge AI applications. However, precise design of diffractive elements (e.g. meta-gratings) requires complex iterative optimization via inverse design algorithms, and the sub-wavelength feature tolerances demand advanced lithography, raising costs and yield issues. Additionally, fixed diffractive patterns limit post-fabrication reconfigurability, constraining adaptability. Integrating tunable materials (e.g., driven by light, electricity, and stress) with micro-actuation systems (e.g., lasers, micro-heaters, and MEMS/NEMS actuators[199]) is a feasible approach to overcome these barriers. For example, laser-defined PCM patterns enable dynamic localized phase modulation of light.[200]

Routing Devices: On-chip routing devices are the "neural wiring" and fan-in/-out of photonic neuromorphic chips, enabling signal routing, power distribution, (de)multiplexing, and optical delays—critical for scalable and reconfigurable PNNs. Large-scale (de)multiplexers—such as high-Q MRR arrays for dense wavelength channels, broadband MMIs, and AWGs—support parallelism but face tradeoffs between channel density and thermal stability. Compact solutions like multi-ring Vernier filters and photonic crystal routers minimize footprints for localized sections. Adaptive directional couplers (DCs), enhanced with tunable materials (e.g. PCMs, $LiNbO_3$), dynamically adjust power splitting to counteract fabrication variations, while ultra-wideband DCs with unconventional structures expand operational bandwidth. For temporal networks like RC, delay elements leverage high-refractive-index lossless materials to compress ns-range delays into sub-mm^2 footprints or exploit microring multiplexers and optical interconnects' intrinsic latency. Innovations such as photonic crystal cavities further promise ultra-compact optical time delay lines via sub-wavelength field confinement.

Future advancements hinge on hybrid integration (PCM-tuned MMIs, $LiNbO_3$ waveguides) and AI-driven self-calibration, transforming passive devices into reconfigurable, area-and energy-efficient components for next-gen optical AI.

5.6. Optical-Electrical Co-Packaging

Photonic integrated circuits (PICs) inevitably rely on peripheral electronics for control and interfacing, including digital-to-analog converters (DACs) for input signal generation, analog-to-digital converters (ADCs) for readout, and ASICs for I/O management and processing. Consequently, any meaningful assessment of photonic computing systems must account for the power consumption of these electronic components alongside the PICs themselves. In practice, the energy overhead of supporting electronics can emerge as a critical bottleneck, potentially offsetting the intrinsic energy efficiency advantages of photonic computing. However, ongoing advancements in semiconductor technology and circuit design are steadily improving electronic energy efficiency. Recent developments include 8-bit 1 GS s^{-1} ADCs achieving a remarkable 1.18 pJ/sample efficiency.[201] Furthermore, heterogeneous integration techniques—such as 2.5D/3D co-packaging of photonic and electronic dies—are showing significant promise for reducing interconnect losses and minimizing driver/sampling power requirements.

Optical-electrical co-packaging addresses the critical bottlenecks of latency, bandwidth, and energy efficiency in modern data systems by directly integrating PICs and electronic integrated circuits (EICs) within a unified package, minimizing interconnect distances (Figure 9). This synergy slashes

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power losses and boosts data rates to Tb/s/mm^2 levels with sub-nanosecond latency, essential for next-gen AI-driven data centers. Current packaging leverages edge coupling (low-loss, high-bandwidth but fragile) and grating coupling (stable, larger size and higher loss) for laser-to-chip interfacing,[202,203] alongside heterogeneous integration of vertical-cavity surface-emitting lasers (VCSELs) tilted for waveguide coupling—enabling compact, high-density links.[202,203] Advanced 2.5D flip-chip methods replace wire bonding with copper microbumps or passive interposers, cutting parasitic capacitance and enabling direct PIC-EIC communication.[204–206] Emerging through-silicon vias (TSVs) and through-oxide vias (TOVs) further shrink footprints but face cost and yield hurdles.[207–209] 3D hybrid integration represents the pinnacle of photonic-electronic co-design, vertically stacking EICs atop of PICs, using TSV, Cu–Cu hybrid bonding, and active photonic interposers, significantly reducing parasitics while enabling >10 k mm^{-2} interconnect density via submicron-pitch microbumps. As the heat generated by EICs can impact PICs, advanced thermal management remains pivotal: integrated TECs stabilize temperature-sensitive photonic components (e.g., microring resonators) and actively dissipate heat from power-dense EICs. While co-packaging unlocks unprecedented performance, challenges in customization flexibility and thermal/mechanical stress demand continued innovation in hybrid integration and materials engineering to achieve scalable, cost-effective solutions.

6. Conclusion

In summary, this review has provided a comprehensive overview of the recent progress in integrated neuromorphic photonic computing, covering advances in linear and nonlinear photonic devices, photonic neural network architectures, and training strategies. We highlighted how PICs unlock ultrafast, energy-efficient matrix operations that fundamentally transcend the limitations of electronic accelerators, positioning photonic neural networks as a transformative paradigm for post-Moore's computing. Our systematic evaluation of critical device innovations—phase-change devices, microring resonators, and III–V neurons—highlights both their performance benefits and remaining scalability hurdles. At the system level, emerging architectures such as photonic CNNs, multilayer perceptrons, diffractive neural networks, spiking neural networks, and reservoir computing demonstrate the versatility of photonic platforms for diverse AI workloads across both cloud-scale and edge scenarios.

Looking ahead, several directions stand out for future research. First, achieving fully integrated, CMOS-compatible on-chip lasers, amplifiers, and high-speed photodetectors will be key to realizing complete PIC-based neuromorphic systems. Second, further exploration of material platforms—including volatile/programmable phase-change materials, and novel 2D materials—will enhance device performance. Third, co-design strategies that tightly integrate hardware with photonic-native algorithms, such as optical backpropagation and in situ training, will accelerate the transition from proof-of-concept demonstrations to large-scale deployment. Finally, advancing packaging and heterogeneous integration will be critical to overcoming fabrication variability and ensuring scalability to wafer-level systems. With sustained multidisciplinary progress, neuromorphic photonic computing is poised to redefine the AI hardware landscape—delivering unprecedented compute density, ultralow power consumption, sub-ns latency, and adaptive intelligence across scales. From hyperscale data centers to autonomous edge devices, PIC technology will play a pivotal role in shaping the next generation of energy-efficient, real-time AI acceleration.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

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